

Table S1: Full five-ratio CIFAR-100 matrix used in the IVC submission. Values are Top-1 accuracy (%) reported as mean $\pm$ std over five seeds.

	$p = 0.5$	$p = 0.6$	$p = 0.7$	$p = 0.8$	$p = 0.9$
<i>ResNet-56</i>					
GRA-CNN	68.79 $\pm$ 0.26	67.47 $\pm$ 0.08	67.76 $\pm$ 0.17	61.49 $\pm$ 0.11	57.51 $\pm$ 0.16
CHIP	68.67 $\pm$ 0.17	67.36 $\pm$ 0.21	67.25 $\pm$ 0.21	60.99 $\pm$ 0.11	56.97 $\pm$ 0.09
L1	68.93 $\pm$ 0.24	67.73 $\pm$ 0.14	67.93 $\pm$ 0.15	61.83 $\pm$ 0.15	57.47 $\pm$ 0.30
Taylor	68.83 $\pm$ 0.11	67.36 $\pm$ 0.15	67.28 $\pm$ 0.24	60.32 $\pm$ 0.19	56.01 $\pm$ 0.37
FPGM	68.69 $\pm$ 0.14	68.02 $\pm$ 0.10	65.73 $\pm$ 0.13	61.11 $\pm$ 0.21	50.36 $\pm$ 0.41
HRank	68.70 $\pm$ 0.19	66.75 $\pm$ 0.09	64.46 $\pm$ 0.17	60.38 $\pm$ 0.20	47.43 $\pm$ 0.89
<i>VGG-16</i>					
GRA-CNN	70.44 $\pm$ 0.43	65.33 $\pm$ 0.96	64.37 $\pm$ 0.35	53.26 $\pm$ 0.45	42.84 $\pm$ 0.40
CHIP	68.28 $\pm$ 0.12	66.18 $\pm$ 0.19	64.43 $\pm$ 0.18	53.80 $\pm$ 0.23	42.69 $\pm$ 0.96
L1	70.27 $\pm$ 0.31	67.30 $\pm$ 0.50	62.81 $\pm$ 0.70	51.55 $\pm$ 0.36	17.26 $\pm$ 22.29
Taylor	68.43 $\pm$ 0.32	64.59 $\pm$ 0.45	61.51 $\pm$ 0.48	50.41 $\pm$ 0.49	42.28 $\pm$ 0.52
FPGM	70.57 $\pm$ 0.16	66.69 $\pm$ 1.14	57.69 $\pm$ 1.06	52.04 $\pm$ 0.75	34.52 $\pm$ 1.71
HRank	69.90 $\pm$ 0.29	66.05 $\pm$ 0.43	59.56 $\pm$ 0.97	50.44 $\pm$ 0.77	35.52 $\pm$ 0.71

Supplementary Material for “GRA-CNN: Semantic-Aware Structured Channel Pruning via Gray Relational Analysis”

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1. Full CIFAR-100 matrix

Table S1 reports the full five-ratio CIFAR-100 results used by the main paper. The complete matrix is retained here to keep the main manuscript focused on the primary operating points and robustness interpretation.

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Table S2: Full paired ResNet-56 residual comparison for GRA-CNN against CHIP and L1 across $p \in \{0.7, 0.8, 0.9\}$.

Baseline	Ratio	Δ (pp)	95% CI	p_{Holm}	Hedges' g
CHIP	$p = 0.7$	+0.51	[+0.42, +0.60]	0.0037	+3.50
CHIP	$p = 0.8$	+0.49	[+0.34, +0.65]	0.0171	+2.09
CHIP	$p = 0.9$	+0.54	[+0.40, +0.66]	0.0100	+2.56
L1	$p = 0.7$	-0.17	[-0.38, -0.01]	0.3605	-0.58
L1	$p = 0.8$	-0.34	[-0.52, -0.13]	0.1115	-1.10
L1	$p = 0.9$	+0.04	[-0.15, +0.21]	0.7452	+0.12

2. Residual paired statistics

Table S2 extends the paired ResNet-56 analysis to the intermediate $p = 0.8$ residual cell. This is included here for completeness because the main manuscript concentrates on the primary operating points.

3. Tiny-ImageNet transfer details

Table S3 provides the full paired transfer summary for the five-ratio Tiny-ImageNet sweep. The complementary ratio curve is now discussed in the main paper.

4. Runtime and deployment diagnostics

5. Dense-ratio robustness

6. Multi-seed sensitivity

Table S3: Formal five-ratio, five-seed Tiny-ImageNet / ResNet-18 transfer sweep for GRA-CNN against five baselines. Each cell reports paired seed-matched accuracy statistics across $p \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$.

Ratio	Baseline	n	Baseline mean	GRA-CNN mean	Δ (pp)	p_t
$p = 0.5$	CHIP	5	61.14 $\pm$ 0.27	60.69 $\pm$ 0.24	-0.45	0.028
	L1	5	60.73 $\pm$ 0.26	60.69 $\pm$ 0.24	-0.04	0.803
	Taylor	5	60.96 $\pm$ 0.20	60.69 $\pm$ 0.24	-0.27	0.231
	FPGM	5	60.88 $\pm$ 0.21	60.69 $\pm$ 0.24	-0.18	0.398
	HRank	5	60.68 $\pm$ 0.18	60.69 $\pm$ 0.24	+0.01	0.871
$p = 0.6$	CHIP	5	60.45 $\pm$ 0.14	60.21 $\pm$ 0.29	-0.24	0.208
	L1	5	60.74 $\pm$ 0.26	60.21 $\pm$ 0.29	-0.53	0.012
	Taylor	5	60.00 $\pm$ 0.12	60.21 $\pm$ 0.29	+0.21	0.137
	FPGM	5	60.56 $\pm$ 0.31	60.21 $\pm$ 0.29	-0.34	0.203
	HRank	5	60.19 $\pm$ 0.25	60.21 $\pm$ 0.29	+0.02	0.910
$p = 0.7$	CHIP	5	56.32 $\pm$ 0.07	56.67 $\pm$ 0.10	+0.35	0.002
	L1	5	56.36 $\pm$ 0.16	56.67 $\pm$ 0.10	+0.31	0.029
	Taylor	5	56.11 $\pm$ 0.15	56.67 $\pm$ 0.10	+0.56	0.002
	FPGM	5	56.50 $\pm$ 0.06	56.67 $\pm$ 0.10	+0.17	0.049
	HRank	5	56.32 $\pm$ 0.16	56.67 $\pm$ 0.10	+0.34	0.006
$p = 0.8$	CHIP	5	59.55 $\pm$ 0.22	59.82 $\pm$ 0.27	+0.27	0.077
	L1	5	59.49 $\pm$ 0.17	59.82 $\pm$ 0.27	+0.33	0.137
	Taylor	5	59.46 $\pm$ 0.37	59.82 $\pm$ 0.27	+0.36	0.114
	FPGM	5	59.46 $\pm$ 0.22	59.82 $\pm$ 0.27	+0.36	0.044
	HRank	5	59.37 $\pm$ 0.16	59.82 $\pm$ 0.27	+0.45	0.018
$p = 0.9$	CHIP	5	50.04 $\pm$ 0.13	49.62 $\pm$ 0.16	-0.42	0.011
	L1	5	49.37 $\pm$ 0.09	49.62 $\pm$ 0.16	+0.24	0.005
	Taylor	5	48.02 $\pm$ 0.17	49.62 $\pm$ 0.16	+1.60	<0.001
	FPGM	5	49.83 $\pm$ 0.19	49.62 $\pm$ 0.16	-0.22	0.029
	HRank	5	46.57 $\pm$ 0.74	49.62 $\pm$ 0.16	+3.05	<0.001

Table S4: Scoring-time profile on the CIFAR-100 main matrix (180 runs, five seeds). Runtime is measured as score computation only (excluding fine-tuning). Relative overhead is reported against CHIP.

Method	Mean score time (s)	Std (s)	Relative to CHIP
L1	0.015	0.025	< 0.001 $\times$
Taylor	0.741	0.559	0.024 $\times$
FPGM	0.094	0.034	0.003 $\times$
CHIP	30.587	16.569	1.00 $\times$
GRA-CNN (Ours)	39.272	21.554	1.28 $\times$
HRank	706.009	746.465	23.08 $\times$

Table S5: Measured device-level change for representative GRA-CNN models on RTX 5090. Positive throughput change and negative latency change indicate improvement relative to the unpruned baseline of the same architecture and dataset.

Model	Ratio	Params ↓	FLOPs ↓	Latency Δ	Throughput Δ
ResNet-56 / CIFAR-100	$p = 0.7$	68.99%	68.26%	+10.36%	-8.65%
ResNet-56 / CIFAR-100	$p = 0.9$	89.24%	88.25%	-6.52%	+8.66%
VGG-16 / CIFAR-100	$p = 0.7$	90.87%	90.84%	-14.44%	+40.70%
VGG-16 / CIFAR-100	$p = 0.9$	98.96%	98.93%	-22.76%	+38.67%
ResNet-18 / Tiny-ImageNet	$p = 0.7$	68.19%	68.92%	-23.63%	+58.28%

Table S6: Dense pruning-ratio extension on CIFAR-100. We add five intermediate ratios ($p = 0.45, 0.55, 0.65, 0.75, 0.85$) and re-evaluate all six methods with five seeds per cell on ResNet-56 and VGG-16. Values are mean $\pm$ std Top-1 accuracy.

Method	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90
<i>ResNet-56</i>										
GRA-CNN (Ours)	69.44 $\pm$ 0.14	68.79 $\pm$ 0.26	68.26 $\pm$ 0.15	67.47 $\pm$ 0.08	66.83 $\pm$ 0.19	67.76 $\pm$ 0.17	63.87 $\pm$ 0.09	61.49 $\pm$ 0.11	58.34 $\pm$ 0.22	57.51 $\pm$ 0.16
CHIP	69.67 $\pm$ 0.19	68.67 $\pm$ 0.17	67.86 $\pm$ 0.18	67.36 $\pm$ 0.21	66.17 $\pm$ 0.06	67.25 $\pm$ 0.21	63.29 $\pm$ 0.11	60.99 $\pm$ 0.11	57.52 $\pm$ 0.19	56.97 $\pm$ 0.09
L1-Norm	69.18 $\pm$ 0.06	68.93 $\pm$ 0.24	68.34 $\pm$ 0.15	67.73 $\pm$ 0.14	66.86 $\pm$ 0.13	67.93 $\pm$ 0.15	63.98 $\pm$ 0.23	61.83 $\pm$ 0.15	58.41 $\pm$ 0.40	57.47 $\pm$ 0.30
Taylor	69.09 $\pm$ 0.07	68.83 $\pm$ 0.11	67.95 $\pm$ 0.09	67.36 $\pm$ 0.15	66.46 $\pm$ 0.13	67.28 $\pm$ 0.24	63.12 $\pm$ 0.23	60.32 $\pm$ 0.19	56.71 $\pm$ 0.15	56.01 $\pm$ 0.37
FPGM	69.81 $\pm$ 0.17	68.69 $\pm$ 0.14	68.47 $\pm$ 0.07	68.02 $\pm$ 0.10	66.88 $\pm$ 0.14	65.73 $\pm$ 0.13	64.27 $\pm$ 0.36	61.11 $\pm$ 0.21	58.09 $\pm$ 0.12	50.36 $\pm$ 0.41
HRank	68.71 $\pm$ 0.09	68.70 $\pm$ 0.19	68.03 $\pm$ 0.17	66.75 $\pm$ 0.09	65.62 $\pm$ 0.12	64.46 $\pm$ 0.17	63.00 $\pm$ 0.21	60.38 $\pm$ 0.20	56.75 $\pm$ 0.44	47.43 $\pm$ 0.89
<i>VGG-16</i>										
GRA-CNN (Ours)	69.36 $\pm$ 0.11	70.44 $\pm$ 0.43	68.01 $\pm$ 0.25	65.33 $\pm$ 0.96	64.04 $\pm$ 0.14	64.37 $\pm$ 0.35	58.42 $\pm$ 0.39	53.26 $\pm$ 0.45	43.56 $\pm$ 0.31	42.84 $\pm$ 0.40
CHIP	69.19 $\pm$ 0.16	68.28 $\pm$ 0.12	67.75 $\pm$ 0.17	66.18 $\pm$ 0.19	64.56 $\pm$ 0.15	64.43 $\pm$ 0.18	58.84 $\pm$ 0.20	53.80 $\pm$ 0.23	46.85 $\pm$ 0.49	42.69 $\pm$ 0.96
L1-Norm	69.62 $\pm$ 0.15	70.27 $\pm$ 0.31	68.62 $\pm$ 0.15	67.30 $\pm$ 0.50	64.88 $\pm$ 0.32	62.81 $\pm$ 0.70	53.73 $\pm$ 0.62	51.55 $\pm$ 0.36	43.19 $\pm$ 0.97	17.26 $\pm$ 22.29
Taylor	69.11 $\pm$ 0.22	68.43 $\pm$ 0.32	67.26 $\pm$ 0.16	64.59 $\pm$ 0.45	60.75 $\pm$ 0.50	61.51 $\pm$ 0.48	42.17 $\pm$ 23.02	50.41 $\pm$ 0.49	44.37 $\pm$ 0.48	42.28 $\pm$ 0.52
FPGM	70.10 $\pm$ 0.23	70.57 $\pm$ 0.16	68.06 $\pm$ 0.19	66.69 $\pm$ 1.14	64.20 $\pm$ 0.38	57.69 $\pm$ 1.06	53.45 $\pm$ 0.51	52.04 $\pm$ 0.75	43.25 $\pm$ 0.66	34.52 $\pm$ 1.71
HRank	68.89 $\pm$ 0.19	69.90 $\pm$ 0.29	67.60 $\pm$ 0.16	66.05 $\pm$ 0.43	64.43 $\pm$ 0.29	59.56 $\pm$ 0.97	54.16 $\pm$ 0.56	50.44 $\pm$ 0.77	43.85 $\pm$ 0.31	35.52 $\pm$ 0.71

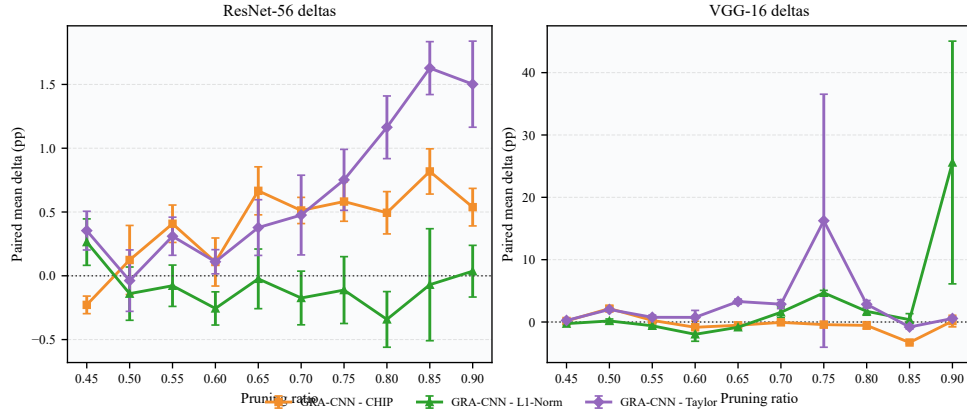


Figure S1: Dense paired deltas of GRA-CNN versus CHIP, L1, and Taylor. The residual-network band is more consistently favorable than the plain-network band.

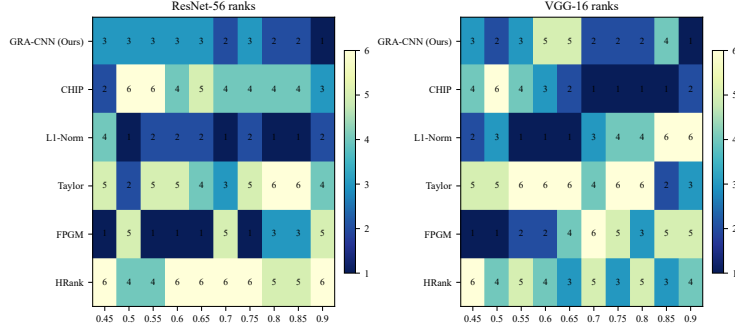


Figure S2: Method rank heatmap across dense ratios. GRA-CNN behaves as a robustness-oriented contender rather than a uniformly dominant score.

Table S7: Extended multi-seed sensitivity of GRA-CNN on ResNet-56 / CIFAR-100 across two compression regimes ($p \in \{0.7, 0.9\}$). Each cell reports mean $\pm$ std over five seeds, together with signal-quality diagnostics derived from the pruning stage.

Ratio	Study	Setting	Top-1 (%)	Score time (s)	Total time (s)	Gate-on	Semantic
$p = 0.7$	ρ sweep	$\rho = 0.1$	65.29 $\pm$ 0.14	16.9 $\pm$ 2.4	284.9 $\pm$ 5.9	0.444	0.444
		$\rho = 0.3$	65.83 $\pm$ 0.26	31.3 $\pm$ 5.9	198.5 $\pm$ 47.3	0.444	0.444
		$\rho = 0.5$	65.72 $\pm$ 0.24	31.6 $\pm$ 5.5	198.7 $\pm$ 48.9	0.444	0.444
		$\rho = 0.7$	65.78 $\pm$ 0.20	32.7 $\pm$ 3.2	200.7 $\pm$ 54.1	0.444	0.444
	Calibration	4 batches	65.75 $\pm$ 0.16	21.5 $\pm$ 2.3	229.5 $\pm$ 86.3	0.444	0.444
		8 batches	65.69 $\pm$ 0.23	28.0 $\pm$ 3.1	170.0 $\pm$ 70.2	0.444	0.444
		16 batches	65.52 $\pm$ 0.11	30.6 $\pm$ 1.6	172.2 $\pm$ 69.0	0.444	0.444
		32 batches	65.38 $\pm$ 0.13	65.2 $\pm$ 21.0	253.6 $\pm$ 70.6	0.444	0.444
	$p = 0.9$	$\rho = 0.1$	51.66 $\pm$ 0.22	33.3 $\pm$ 0.8	207.6 $\pm$ 0.4	0.444	0.444
		$\rho = 0.3$	51.66 $\pm$ 0.22	32.5 $\pm$ 0.4	204.6 $\pm$ 5.9	0.444	0.444
		$\rho = 0.5$	51.66 $\pm$ 0.22	31.4 $\pm$ 0.3	192.5 $\pm$ 0.4	0.444	0.444
		$\rho = 0.7$	51.66 $\pm$ 0.22	31.5 $\pm$ 0.2	192.2 $\pm$ 0.2	0.444	0.444
		4 batches	51.47 $\pm$ 0.09	26.0 $\pm$ 1.7	171.7 $\pm$ 1.3	0.444	0.444
		8 batches	51.72 $\pm$ 0.26	26.3 $\pm$ 0.6	171.5 $\pm$ 0.7	0.444	0.444
		16 batches	51.65 $\pm$ 0.19	30.4 $\pm$ 2.2	172.3 $\pm$ 6.2	0.444	0.444
		32 batches	51.77 $\pm$ 0.17	57.9 $\pm$ 5.5	224.5 $\pm$ 5.2	0.444	0.444

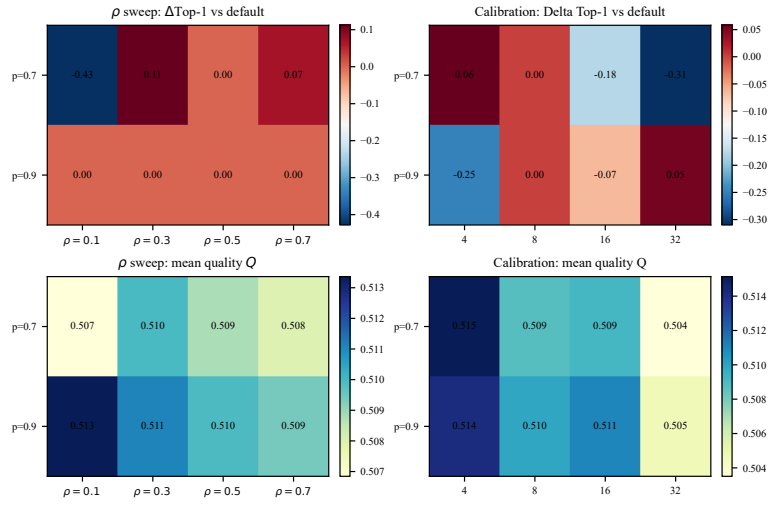


Figure S3: Sensitivity heatmap across study type, compression regime, and setting. The dominant observation is bounded variation rather than sharp tuning sensitivity.